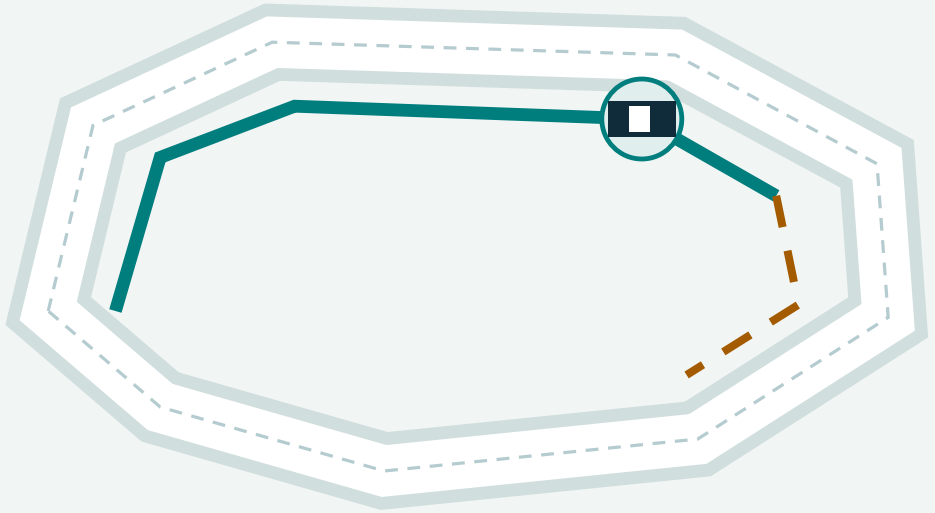
Vehicle localization from the cameras you already run

Seven calibrated cameras, a reference map and motion signals produce one shared vehicle pose.

When geometric evidence is insufficient, the output is unavailable.

SYNTHETIC ILLUSTRATIVE GEOMETRY



One pose. An explicit unavailable state.

POSE AVAILABLE	UNAVAILABLE
`x  y  heading`	No pose published

CUSTOMER DECISION

An additional position signal for a mapped course

Vehicle autonomy leads and track operators

When the existing position signal weakens, can camera evidence provide a useful second input?

THE PILOT DECISION

Is the signal accurate and available often enough for our course and operating conditions?

Existing inputs

Compatible calibrated cameras and a reference map

Proposed integration

A timestamped pose and explicit availability status

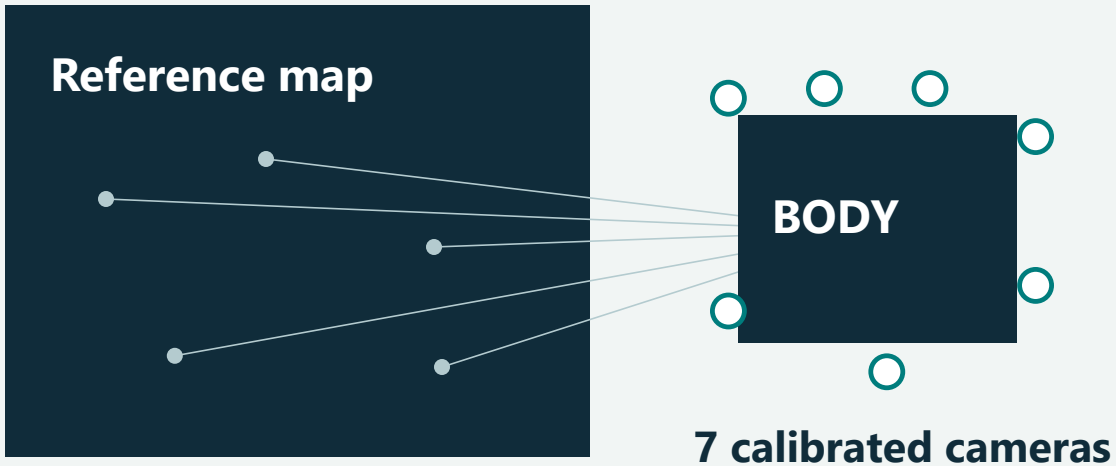
Value to test

Useful supplementary coverage within the existing stack

SOLUTION

One vehicle pose from map geometry and motion

SYNTHETIC ILLUSTRATIVE GEOMETRY



7 calibrated cameras
1 shared body pose

Map correspondences constrain camera poses.

Calibration ties cameras to one vehicle frame.
The quality threshold remains uncalibrated.

CAUSAL ESTIMATION

01 Geometry

Match the map and solve a consistent body pose.

02 Motion

Fuse motion signals with current and past evidence.

03 Quality gate

Publish an accepted pose or mark it unavailable.

VERIFIED AGGREGATE EVIDENCE

Same-session development meets aggregate targets

Laps 6 + 7 / One continuous run / One initial truth seed / No lap 7 reset

17.48cm

Position RMSE

Target ≤20cm

11.42cm

Lateral RMSE

Target ≤20cm

94.8%

One-second coverage

Target ≥90%

Metric	Observed	How to read it
Position error, p95	31.95cm	RMSE is not a per-output guarantee
Within 20cm, all planned	71.24%	327/459 non-seed frames, unavailable = miss
Published / unavailable	411 / 49	460 planned, 89.35% publication fraction

Coverage counts occupied one-second bins. Development data is consumed, not untouched validation.

Coverage and cold start remain the deployment blockers

Original 131-second stress window

18.48 / 9.95cm

Position / lateral RMSE on published poses

72% coverage: FAIL

179/263 published, 84 unavailable.
Separate seeded window. Below 90% target.

Blind day replay

0 / 148

Published / planned groups, all unavailable

Never initialized

No truth labels or initial seed. No RMSE.
No approved three-camera fix.

Next proof: independent day/night runs, reliable cold starts and calibrated confidence.
End-to-end latency is unproven. No production deployment or safety certification.

PROPOSED CUSTOMER PILOT

One vehicle. One mapped course. A decision in four weeks.

Offline replay, then shadow operation

Use compatible existing cameras and recorded motion signals. Keep the output outside vehicle control.

Customer contribution

Course access, calibration, an independent evaluation capture and an agreed latency budget.

Acceptance item	Proposed target
Independent day + night	Freeze map and parameters before capture
Position / lateral RMSE	≤20cm each on published non-seed poses
One-second coverage	≥90% in each agreed test condition
Cold start, no truth seed	≥95% of agreed starts within 30s
Tail error + missing poses	Report p95 and all-planned within 20cm
Signal integrity + latency	100% explicit unavailable states, measure p95

Decision: extend the pilot only if the agreed gates pass.